

3.5.3 Speed and stopping distances

Students should have knowledge and understanding of the following content.

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Outcome 6			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
Speed is measured by the distance travelled in a certain time. Units of speed include metres per second and kilometres per hour. Simple calculations of average speed using the equation: $\text{speed} = \text{distance}/\text{time}$ will be required.	Suggested activity for TDA Investigate how the speed of a trolley changes as it rolls down a slope.	6.5.4.1.2	4.7.1.1
Outcome 7			
The stopping distance of a vehicle is the sum of the distance the vehicle travels during the driver's reaction time (thinking distance) and the distance it travels under the braking force (braking distance). For a given braking force the greater the speed of the vehicle, the greater the stopping distance.	Students may find it helpful to refer to the Highway Code.	6.5.4.3.1	4.7.1.10
Outcome 8			
Reaction times vary from person to person. Typical values range from 0.2 s to 0.9 s. Knowledge and understanding of methods used to measure human reaction times. Knowledge of how a driver's reaction time can be affected by tiredness, drugs and alcohol. Distractions may also affect a driver's ability to react.	Students should be able to interpret and evaluate measurements from simple methods to measure the different reaction times of students. Students should be able to evaluate the effect of various factors on thinking distance. Suggested activity for TDA Investigate factors that affect human reaction time, eg tiredness, distraction, practise.	6.5.4.3.2	4.2.1.6
Outcome 9			

Content	Additional guidance and suggested TDAs	Specification reference GCSE Combined Science: Trilogy	Specification reference GCSE Combined Science: Synergy
The braking distance of a vehicle can be affected by adverse road and weather conditions, and poor condition of the vehicle. Students should be able to analyse a given situation to identify how braking could be affected.	Adverse road conditions include wet or icy conditions. Poor condition of the vehicle is limited to the vehicle's brakes or tyres.	6.5.4.3.3	4.7.1.10